

WP HERMES

High fidelity numerical simulations of strongly coupled processes
for repository Systems

QUESTIONNAIRE FOR POTENTIAL CONTRIBUTORS TO WP-HERMES

We launch this questionnaire to further develop the work package description where your organization has expressed interest. This Questionnaire will serve as the input for the WP-HERMES Work Package “High fidelity numerical simulations of strongly coupled processes for repository Systems”. The information requested include 4 part:

- 1) Partner Information
- 2) Possible role of your institution in WP-HERMES
- 3) Description of the available expertise and infrastructure
- 4) Description of potential work in WP-HERMES

The reference document for the WP-HERMES scope & budget is the Template #2 approved by the core group and colleges

The selection of the partners will be based in the information provided in the questionnaire. Please be accurate, clear and concise. The main criteria for the selection process are the scientific excellence, relevance, cost/benefits positive attitude towards the collaborative spirit of joint EURAD program.

Please download this questionnaire and budget table to

<https://drive.switch.ch/index.php/s/DYcO4G1j3nLkSJ3> (DropBox). Access code: HERMES

by the August 17th, 2023!

Project description (Template #2) and the budget table is available at:

<https://drive.switch.ch/index.php/s/Hr6db5FUeAYhAfa>

Project specific question can be send to dedicated e-mail: WP18HERMES@gmx.ch;

PARTNER INFORMATION

Partner Acronym	Country	College	Level of interest* H/M/L	Willingness to join End-user Group**
BGR	Germany	RE	M	N

*L = Low: Supportive, will likely use the results from the WP (e.g. as an end-user)

M = Medium: Supportive, will likely contribute to the WP (e.g. contribute effort/time)

H = High: Supportive, will likely play a central role in the WP (e.g. Task leader)

** The duties of the End-user group are: Evaluation of work program in terms of scope, excellence and implementation/safety case relevance. SoTA: contribution and review. End-user group can include partners from RE/TSO/WMO. Work load ~ 1 PM.

	Name	Institute	Email
Person completing survey			
Contact Person (if different)	Gesa Ziefle	BGR (Geotechnical Safety Analysis)	Gesa.Ziefle@bgr.de
Other partner organisations involved in the experiment, if any.	Olaf Kolditz	UFZ	Olaf.kolditz@ufz.de
	Thomas Nagel	TUBAF	thomas.nagel@ifgt.tu-freiberg.de
	Michal Béres	UGN	michal.beres@ugn.cas.cz
	Nicolas Tardieu	EDF	nicolas.tardieu@edf.fr

Note: If your interest is only limited to be an end-

user group than you have finished with completing this questionnaire.

POSSIBLE INVOLVEMENT IN WP

Please select task(s) you would like to participate in (based on *WP description in template #2*:

Task 2: Knowledge Management • SOTA report(s) & Knowledge gap analysis • Implementation and professional maintenance of collaborative platform for code development and machine learning. • Implementation of EURAD model-hub for linking data and models. • Maintenance of open access platform with experimental data for model testing	☐ Expected budget: ... [KEUR] Expected effort: ... [PM]
Task 3: Process couplings and computational performance • Definition and development of unified interfaces for process coupling • Development and coupling of tools for forward and inverse modelling as well as sensitivity analysis • Development of models with dynamic integration of data • Multiscale models couplings, parameters transfer and upscaling	☐ Expected budget: ... [KEUR] Expected effort: ... [PM]
Task 4: Surrogate modes of individual phenomena • Processes addressed: Chemistry / Gas-Mass-Heat transport / Mechanics • Application & implementation of machine learning methods and codes • Benchmarking of surrogate models against physical models and experimental data	☐ Expected budget: ... [KEUR] Expected effort: ... [PM]
Task 5: Tailored models for SA/PA and field scale mock-ups • Real time simulation of field scale experiments • Assessment of waste package integrity in extended storage • High fidelity models for repository near-field simulations • Dedicated models for optimisation and repository design • Integration of the models into DT concepts to be formalised in WP17	☒ Expected budget: 38.7 [KEUR] Expected effort: 1.88 [PM]

EXPERTISE AND TOOLS AVAILABLE

Experience in the fields relevant for WP-HERMES	BGR has decades of experience in the implementation, development, and evaluation of numerical modeling methods for coupled THM processes, especially in the context of final disposal. This has been gained in international (DECOVALEX, Mont Terri, EBS Task Force, EURAD I, BEACON...) as well as national projects, and participation in safety investigations on the Asse and Morsleben sites. Within the current DECOVALEX phase and EURAD Gas, BGR is involved in SA/PA tasks; in the national projects VSU, ANSICHT, CHRISTA, KOSINA and RESUS, BGR together with BGE-TECHNOLOGY and GRS was/is involved in the development and exemplary application of a concept for the safety assessment of possible repository systems in clay, salt and crystalline rock in Germany, partly including UQ. Model validation is performed, among others, by numerical simulation of near-field processes in large-scale experiments, such as Mont Terri (CD-A, FE, Sandwich, FS), ÄSPÖ, Bure, Horonobe.
Link with other WP within EURAD-I	Links exist with the current WP GAS (EURAD I).
Key relevant publications (max 5)	Ziefle, G., Cajuhi, T., Graebing, N., Jaeggi, D., Kolditz, O., Kunz, H., Maßmann, J., Rink, K. (2022): Multi-disciplinary investigation of the hydraulic-mechanically driven convergence behavior: CD-A twin niches in the Mont Terri Rock Laboratory during the first year. <i>Geomech. Energy Environ.</i> 31 , art. 100325 Sci. 82 (1), art. 42 Ziefle, G., Cajuhi, T., Costabel, S., Furche, M., and Maßmann, J. (2023): Water content evolution in the EDZ of Opalinus Clay: A methodic approach for a comparative interpretation of measurements and modelling. <i>Rock Mechanics and Rock Engineering</i>, submitted Jobmann, M., Bebiolka, A., Burlaka, V., Herold, P., Jahn, S., Lommerzheim, A., Maßmann, J., Meleshyn, A., Mrugalla, S., Reinhold, K., Rübel, A., Stark, L. & Ziefle, G. (2017): Safety assessment methodology for a German high-level waste repository in clay formations. <i>Journal of Rock Mechanics and Geotechnical Engineering</i>, 9, 5: 856-876. DOI:10.1016/j.jrmge.2017.05.007 Maßmann, J., Thiedau, J., Bittens, M., Kumar, V., Tran, T.V., Morel, C.G., Kneuker, T. & Schumacher, S. (2022): ANSICHT-II – Methode und Berechnungen zur Integritätsanalyse der geologischen Barriere für ein generisches Endlagersystem im Tongestein. Bundesanstalt für Geowissenschaften und Rohstoffe (BGR), Ergebnisbericht; Hannover. DOI:10.25928/n8ac-y452

Available computational infrastructure	4 conventional Linux-Server with up to 100 cores 2 KI-Server, each with 4 NVIDIA Tesla V100 GPU
Availability or / access to experimental data for code validation? (Describe)	BGR is the principle investigator in the Mont Terri CD-A-Experiment. This long term in-situ experiment focuses on the coupled hydraulic and mechanical behaviour of the Opalinus clay around two niches. BGR has access to long-term, ongoing measurements on deformation, humidity, water content, pore water pressures, crack width, etc. There exist pointwise measurements (as piezometer, extensometer) and two-dimensional measurements, as ERT, TDR and laser scans. Especially for the water content evolution different results from different measurement devices are available. Furthermore, geological characterization of inhomogeneities, as fractured zones, are available.
Available / maintained modelling code	(name/type/parallelisation/couplings/...) BGR uses the OpenSource FE code OpenGeoSys (OGS). In collaboration with the UFZ, BGR is also involved in the further development of the code. BGR is experienced in parallel computing of large systems with THM coupling, considering two-phase flow.
PhD/PostDoc planned in the field of HERMES	The participation of PhD/PostDoc in the planned work is being sought.

Modelling tools

Modelling codes available (name)	Type (FE,DEM, MM,FV,...)	Scales covered (Field, Continuum, Pore molecular, ...)	Opensource code? (Y/N), code developed within EURAD? (Y/N)
OpenGeoSys 6	FE	Field to lab scale, based on mixing theory of porous media	Y Y (partly)

Process Couplings to be considered

	Thermal	Mechanical	Chemical	Hydraulic
Thermal	☒			
Mechanical	☒	☒		
Chemical	☐	☐	☐	
Hydraulic	☒	☒	☐	☒

PROPOSED WORK

(Max 100 words per Task2; Max 300 words Tasks 3-5)

T2	
T3	
T4	
T5	The work package aims at simulating strongly coupled effects in the near-field of repository systems. Based on this, the interpretation of field experiments is envisaged. An application for data-driven modelling based on measurements will be the CD-A experiment at Mont Terri (Ziefle et al. 2022). Based on long term measurements and observations, machine learning (ML) methods could be used to analyse the EDZ and the associated water content evolution, with the aim of a process analysis and associated parameterisation of the EDZ as a basis for classical process modelling. This type of measurement driven modelling is an important complement to ML (e.g. neural networks) as it allows heterogeneities to be taken into account and will thus also be an important basis for classical process modelling. The planned Model-Hub (see UFZ Task 2) for the combined use of different codes and related data sets would be a valuable tool and important basis for such modelling approaches. Links exist with the other activities such as the Mont Terri project. The aims of this work could be summarized as follows: • Improved understanding of EDZ evolution (TH2M) • Workflow for measurement-driven modelling • Validation on field scale (CD-A experiment)

Explain how proposed work is different/similar from current activities in EURAD-I (200 words max)

The planned work is based on the experience gained from EURAD-I in the context of numerical modelling for the safety assessment of potential repository systems in claystone with OGS. However, EURAD II will focus on two aspects: a) The development of strategies for the effective modeling of heterogeneous near-field processes in the clay rock under inclusion of measurement data at the large-scale in-situ experiment CD-A. A modelling concept considering measurement driven approaches (e.g. ML) will be conceptualized in collaboration with the team from UGN (CZ), in particular to take into account the influence of inhomogeneities as encountered in the EDZ. b) The validation of numerical modelling approaches on the basis of in-situ measurements is being further advanced in collaboration with the team from EDF (FR) using the example of the CD-A experiment in Mont Terri.

Indicates risks accompanying implementation of proposed work

The risks are considered to be very low, as a lot of experience already exists. One risk is that not all targets can be achieved in the planned time. This could be caused by capacity constraints, especially in terms of personnel.

The proposed development of strategies for near field effects is based on the existence of a related data set. This data set already exists for a time period of about 5 years.

Consequently, the risk of a lack of data is very low. The proposed concept is based on promising approaches, but parts of it are new and it will only become clear in the course of the work how effectively it can be applied to problems in practice.

Define 3 possible KPI and criteria of project success for evaluation after 48 month

- Results from a 3D THM model of the CD-A-experiment are compared with measurements
- A concept for measurement-driven process-modelling is defined
- This concept was applied as an example to selected measurements

Possible links with other EURAD-II projects qualified for template#3

Links exist with the planned WP17 Digital Twin (DITOCO) and WP12 Bentonite (ANCHORS).